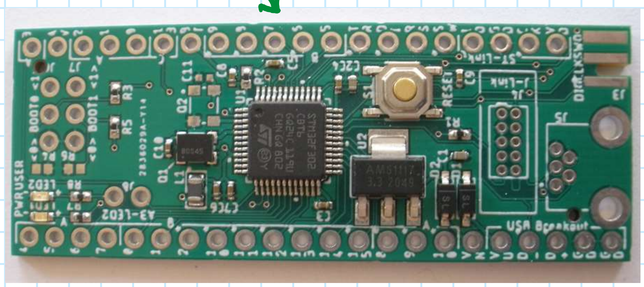
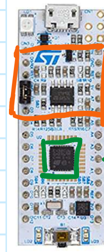
Make Your Own MCU Boards

Thursday, June 1, 2023 2:51 PM

The Goal

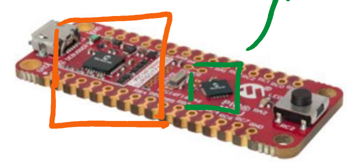
STM32F302





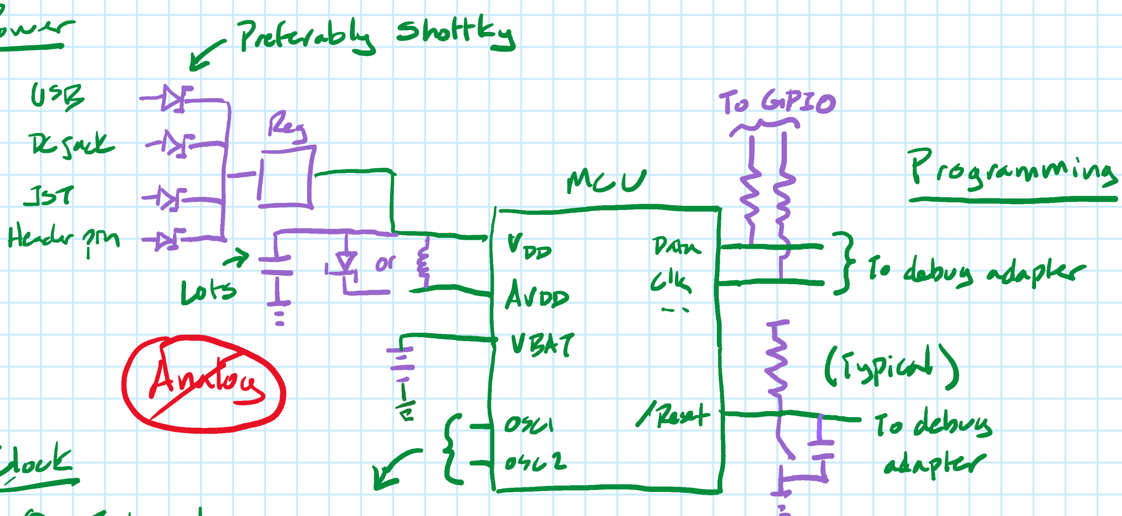
"PICkit"

PIC16F18446



① Draw the circuit → Check the MCU pinout & Hardware development guide

Power

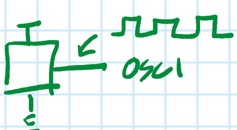


Clock

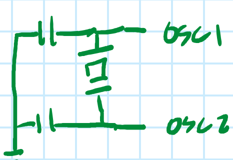
① Internal

↳ ~2-5% / Sometimes trimmable

② Oscillator



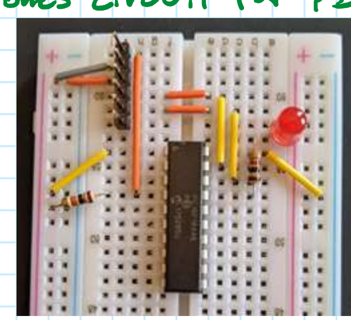
~~③~~ Crystal/Resonator



Misc

- Boot pins
- Unused pins / low power
- Power LED, LED_BUILTIN, BTNA

Ex bare-bones circuit for PIC16:



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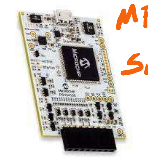
Don't forget the debug adapter!

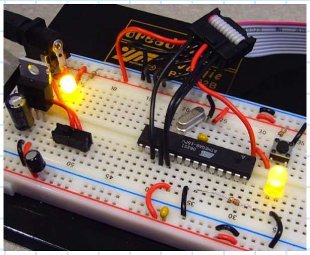
ST-Link V2

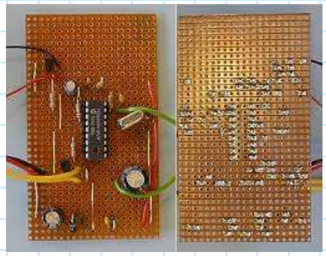
V3

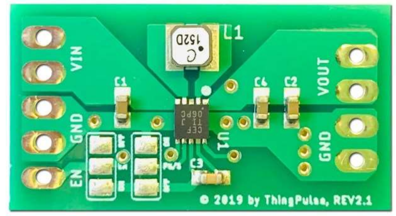
MPLAB Snap



② Choose parts



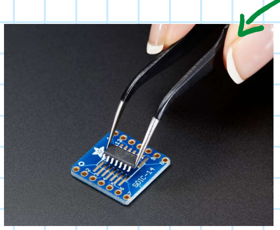




Through-hole/Breakout

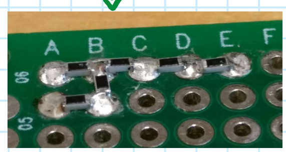
Whatever you can solder!

PCBA parts library

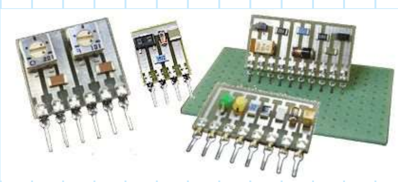


Adafruit #1210

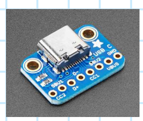
Large-ish SMT (0805, 1206, etc)



jlcpcb.com/parts

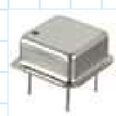


capitaladvanced.com/6000ser.htm

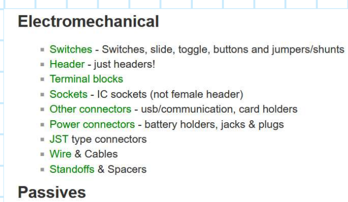


Adafruit #4090

Adafruit #373

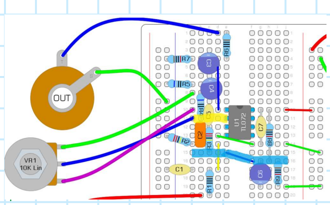


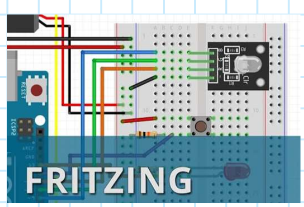
Digi-Key X209-ND

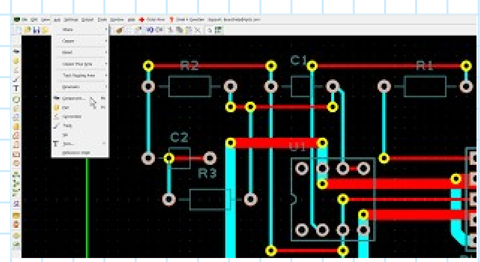


ladyada.net/wiki/partselector

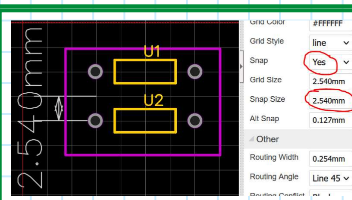
③ Layout the circuit







EasyEDA →



CAD w/ 0.1" grid

Max freq?

~ 1s of MHz

① > 50 MHz

② $> \dfrac{6.1}{0.012 \cdot L}$ MHz *

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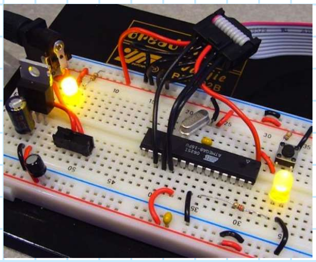
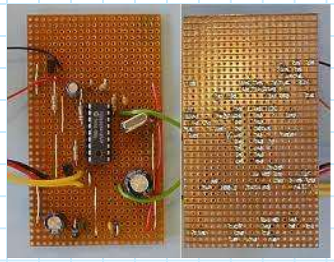
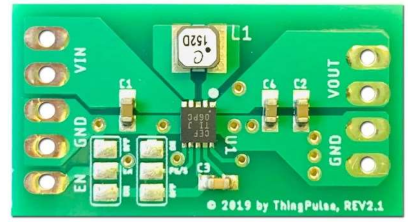
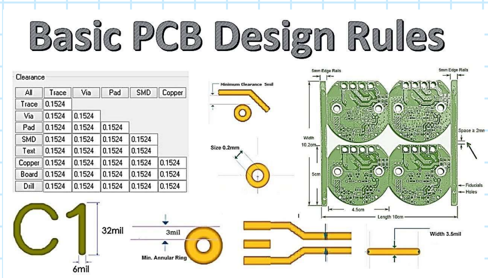
Friday, June 9, 2023 7:41 AM

③ Layout the circuit (cont)



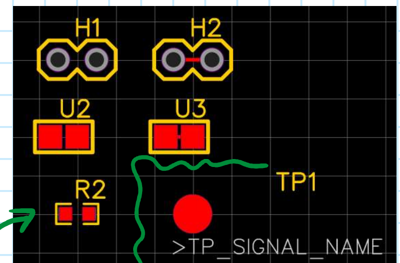




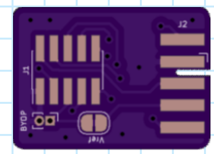


github.com/JLCPCB/jlcpcb-eagle
jlcpcb.com/capabilities/Capabilities

NO NC



Populate w/ 0Ω to close

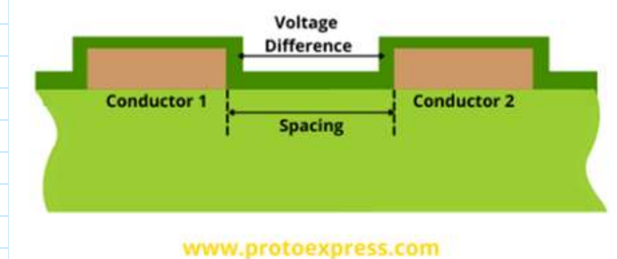


debug-edge.io

Max I? V?

10 mil trace, exterior = 3.5A

Closest traces 24 mil = 100 VDC
(headers @ 0.1" spacing)

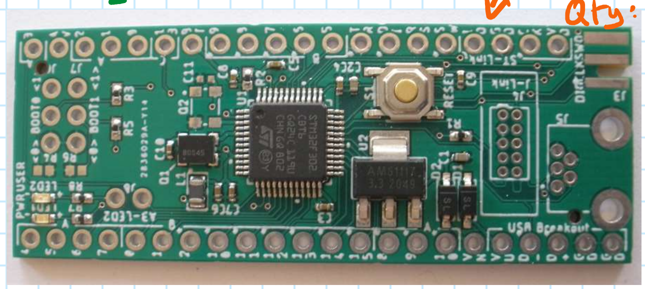


④ Construction

JLC PCB Assembly

$ = 9.5 + 0.0017 · SMT + [3/ext part]
+ [3.5 + 0.0173 · PTH]

$3-5 @ Qty: 10



Maker Fabs

$ = 28 + 0.03 · SMT + 0.04 · PTH